ADDITIONAL DOCUMENTATION FOR DATASET UNDERSTANDING

1. Dataset Overview

This dataset was created to support a reinforcement learning (RL) malware investigation framework. It contains comprehensive memory snapshots from various Windows operating system environments, both clean and infected with different malware samples. The dataset enables researchers to study malware behaviours and systemic changes in memory under controlled conditions.

2. Dataset Composition

- **Operating Systems:**

  The dataset includes 13 different Windows OS versions, including but not limited to:

  - Windows 7 Professional (multiple editions and service packs)
  - Windows 8.1 Pro (32-bit and 64-bit)
  - Windows 10 (various updates, 32-bit and 64-bit)
  - Windows 11 22H2 (64-bit)
  - Developer and evaluation editions (WinDev2303Eval)

- **Malware Samples:**

  13 distinct malware types were used to infect the systems, chosen to represent different attack vectors and behaviours:

  - WannaCry, Cerber, DeriaLock, $LuckyLcoker, Dharma, SporaRansomware, GandCrab, GoldenEye, Locky.AZ, InfinityCrypt, Win32.BlackWorm, PowerLoader, W32.MyDoom.A.

- **Data Files:**

  We took two memory snapshots, one uninfected and one infected, for each OS version. This results in a total of 26 RAM dump files (13 clean and 13 infected).

3. Data Collection Process

- Experiments were conducted in a controlled virtual machine environment isolated on a secure network at London Metropolitan University Digital Forensics Laboratory.
- Each OS was installed from ISO files, and initial system snapshots were captured.
- Malware infection was then applied, followed by capturing a second snapshot.
- Memory dumps were extracted and analysed using the Volatility framework, a tool for memory forensics.
- Manual examination of the infected and clean memory snapshots helped identify changes and patterns indicative of malware presence.

4. Data Structure

Each sample consists of:

- A raw memory dump file (.raw or .bin format) representing the entire RAM of the virtual machine.
- The sample also includes metadata that describes the OS version, the infection status (clean or infected), and the type of malware (if infected)
- The experiment details and configuration for each sample are detailed in supplementary documentation.

5. Instructions for Use

- Researchers should first familiarise themselves with the OS versions and infection status metadata for each file.
- Use memory forensics tools, such as Volatility, to analyse RAM dumps. Volatility plugins can extract processes, network connections, loaded modules, and other artefacts.
- Compare clean and infected memory snapshots for the same OS to identify malware indicators.
- Refer to the provided malware workflow diagram (included in the documentation) to understand typical malware behaviours and the expected system impact.
- Use the dataset to train, validate, or benchmark malware detection or investigation frameworks, especially those employing reinforcement learning techniques.

6. Additional Notes

- The isolated environment ensures no risk of spreading malware during analysis.
- All experimental procedures are documented to enable reproducibility.
- This dataset offers a unique combination of multiple OS versions and diverse malware types, useful for broad-spectrum malware research.

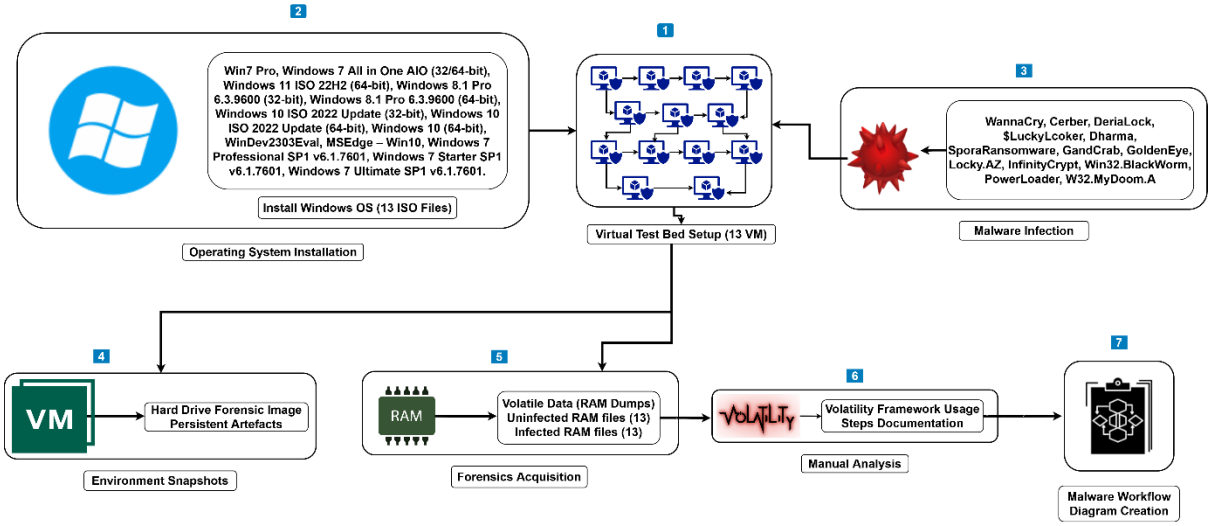


Fig 1: Malware Workflow Diagram

1. WannaCry-Win7 Pro Spanish	05/12/2023 17:14	File folder
2. Cerber - Windows 7 All in One AIO (3264-bit)	05/12/2023 17:16	File folder
3. DeriaLock - Windows 11 ISO 22H2 - 64bit	05/12/2023 17:19	File folder
4. $LuckyLcoker - Windows 8.1 Pro 6.3.9600 - 64bit	05/12/2023 17:23	File folder
5. Ramsomware - Dharma - Windows 8.1 Pro 6.3.9600 - 32bit	05/12/2023 17:26	File folder
6. Ransomeware - SporaRansomware - Windows 10 ISO 2022 Update 32bit	05/12/2023 17:29	File folder
7. Ransomeware - GandCrab - Windows 10 ISO 2022 Update-64bit	05/12/2023 17:33	File folder
8. Ransomeware - GoldenEye - Windows 10-64bit	05/12/2023 17:36	File folder
9. Ransomeware - Locky.AZ - WinDev2303Eval	05/12/2023 17:39	File folder
10. Ransomeware - InfinityCrypt - MSEdge - Win10	05/12/2023 17:43	File folder
11. Malware - Win32.BlackWorm - Windows 7 Professional SP1 v6.1.7601	05/12/2023 17:46	File folder
12. Malware - PowerLoader - Windows 7 Starter SP1 v6.1.7601	05/12/2023 17:47	File folder
13. Malware - W32.MyDoom.A. (or_and) ShadowBot - Windows 7 Ultimate SP1 v6.1.7601	05/12/2023 17:12	File folder

Figure 1: Creating a Robust Malware Dataset for the RL Framework

$LuckyLcoker - NoVirus.mem	$LuckyLcoker - WithVirus.mem	Cerber - NoVirus.mem
Cerber - WithVirus.mem	DeriaLock - NoVirus.raw	DeriaLock - WithVirus.mem
Dharma - NoVirus.mem	Dharma - WithVirus.mem	GandCrab - NoVirus.raw
GandCrab - WithVirus.raw	GoldenEye - NoVirus.raw	GoldenEye - WithVirus.raw
InfinityCrypt - NoVirus.raw	InfinityCrypt - WithVirus.raw	Locky.AZ - NoVirus.mem
Locky.AZ - WithVirus.mem	PowerLoader - NoVirus.mem	PowerLoader - WithVirus.mem
SporaRansomware -NoVirus.mem	SporaRansomware -WithVirus.mem	W32.MyDoom.A. - NoVirus.mem
W32.MyDoom.A. - WithVirus.mem	WannaCry - NoVirus.raw	WannaCry - WithVirus.raw
Win32.BlackWorm -NoVirus.mem	Win32.BlackWorm -WithVirus.mem	

Figure 2: The 26 Robust Malware Datasets for the RL Framework